

Lab 02 Strings

Objective:

In this lab, you will learn how to:

- Declare and manipulate data of the char data type.
- Write string processing program using String, StringBuilder, and StringBuffer objects.

Activity Description:

(**Unique Course ID**): Most of Learning Management Systems (such as Blackboard) keep a record for all given courses during each semester for that those systems need to have a unique ID for each course which composed of: course code, course number, year, semester and Section Reference Number.

Write a program that help to generate and print a unique ID for a course by prompting the user to enter course information while checking if they are matching the following rules:

1. **Course Code** (String): consist of only capital letters with a length between 2 and 4
2. **Course Number** (String): consist of only 3 digits
3. **Year** (int)
4. **Semester** (int): Fall (01), Spring (02) and Summer (03)
5. **Section Reference Number** (int)

After checking each piece of data add its value to **CourseID** (StringBuilder) following the same format as in the sample output

Sample Output:

Sample(1)

```
Course ID Generator:
-----
Please Enter Course Information
Course Code (only capital letters with a length between 2 and 4):CPCS
Course Number (consist only of 3 digits):203
Year:2019
Semester (Fall=01, Sprint=02, Summer=03):2
Course Reference Number:12345
-----
Generated Course ID: CPCS203-201902-12345
```

Sample(2)

Course ID Generator:

Please Enter Course Information

Course Code (only capital letters with a length between 2 and 4):c

Course Code length was not between 2 and 4 , Please try again

Course Code (only capital letters with a length between 2 and 4):cpcs

Course Code must only have capital letters , Please try again

Course Code (only capital letters with a length between 2 and 4):CPCS

Course Number (consist only of 3 digits):20

Course Number length was not 3 , Please try again

Course Number (consist only of 3 digits):203

Year:2019

Semester (Fall=01, Sprint=02, Summer=03):4

Please enter correct semester code, try again

Semester (Fall=01, Sprint=02, Summer=03):3

Course Reference Number:12345

Genereated Course ID: CPCS203-201903-12345